Tribunal — Tech Stack Doc

Constraint

Zero-cost MVP. Every tool below must have a sustainable free tier for development and demo usage — not a trial that expires.

Stack

Layer	Choice	Notes
Backend framework	FastAPI (Python)	Native async support — needed for parallel agent calls (Proponent + Antagonist run concurrently in v1+)
Backend hosting	Railway or Render (free tier)	Free tier sleeps on inactivity — acceptable for dev/demo, ping before live demo or run locally
Frontend	Next.js + React + Tailwind	Standard, free, well-documented
Frontend hosting	Vercel (free tier)	Generous limits, no cold-start issue unlike backend free tiers
Database	PostgreSQL via Supabase or Neon (free tier), or local Docker for dev	Not yet wired into v0 — needed once run history/persistence is added
Primary LLM	Google Gemini API (free tier)	Best free-tier ceiling among frontier-adjacent models as of this build
Secondary LLM	Groq (free tier)	Fast inference, open models — backup/overflow when Gemini rate-limits, or for provider diversity in blind-draft stage

Layer	Choice	Notes
Verification (v1, stretch)	Tavily API (free tier)	Limited free searches/month — budget carefully once wired in
Version control	GitHub	Standard

Model Versioning Note

Model names change (`gemini-2.0-flash` was deprecated mid-build, replaced by `gemini-3.6-flash`). Pin the model name in one place — an environment variable or a single config constant — not hardcoded across multiple files, so future deprecations are a one-line fix instead of a search-and-replace across the codebase.

Known Constraints of Free-Tier-Only Stack

- **Rate limits:** each team member should use their own API key during development, not share one, to avoid silent throttling that gets misattributed to "the model is broken"
- **Cold starts:** free-tier backend hosts sleep after inactivity; factor this into demo-day planning (warm it up beforehand, or run locally for the actual demo)
- **No sustained high-volume testing:** free tiers are fine for individual dev/test cycles, not for load testing — that's out of scope for an MVP anyway

What's Deliberately Not in the Stack (v0)

- No paid LLM APIs (OpenAI/Anthropic direct) unless the team has unspent credits — not assumed as available
- No SSE/WebSocket infrastructure yet — v0 uses simple request/response, not streaming
- No queue/task system (Celery, etc.) — v0's call volume doesn't need it; revisit if v1 fact-checking introduces long-running batch jobs